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REVIEW

Sex differences in telomeres and lifespan

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Summary

Males and females often age at different rates resulting in longevity 'gender gaps', where one sex outlives the other. Why the sexes have different lifespans is an age-old question, still fiercely debated today. One cellular process related to lifespan, which is known to differ according to sex, is the rate at which the protective telomere chromosome caps are lost. In humans, men have shorter lifespans and greater telomere shortening. This has led to speculation in the medical literature that sex-specific telomere shortening is one cause of sex-specific mortality. However, telomere shortening may be a cause for and/or a consequence of the processes that govern survival, and to infer general principles from single-taxon studies may be misleading. Here, we review recent work on telomeres in a variety of animal taxa, including those with reverse sexual lifespan dimorphism (i.e., where males live longer), to establish whether sex-specific survival is generally associated with sex differences in telomere dynamics. By doing this, we attempt to tease apart the potential underlying causes for sex differences in telomere lengths in humans and highlight targets for future research across all taxa.

Key words: gender; heterogamety; hormones; lifespan; oxidative stress; telomere dynamics; sexual size dimorphism.

Telomeres and lifespan

Vertebrate telomeres are made of non-coding TTAGGG DNA repeats that, along with associated proteins, cap and protect the terminal ends of linear chromosomes, preventing chromosomal fusion, degradation of coding DNA, and the ends being recognized as breaks (Blackburn, 2000). Telomeric base pairs can be lost through processes including incomplete DNA replication ('end replication problem'; Olovnikov, 1996) and damage from agents such as reactive oxygen species (ROS hereafter; von Zglinicki, 2002). A number of telomere restoration processes also occur, most notably via the action of the enzyme telomerase, which adds base pairs *de novo* (Greider & Blackburn, 1989). However, telomerase down-regulation is common in the adult somatic tissues of large and/or long-lived mammals, including humans (Gomes *et al.*, 2011). Because critically short telomeres induce cellular replicative senescence and/or cell death (Campisi, 2005), by removing telomerase-driven maintenance, telomere shortening limits the number of times a cell may divide and live. Hence, telomerase down-regulation may have evolved in response to the increased risk of cell immortalization and cancer that accompany larger size (i.e., more cells) and/or long lifespan. Without telomerase, it is logical that telomeres will shorten with age; however, age-dependent telomere shortening is also found in some species that do not down-regulate telomerase in adult somatic tissue (Tables 1 and S1); this suggests that the chronic actions of telomere shortening outweigh those of telomere maintenance in a variety of species.

Both short telomere size and increased attrition rate are associated with reduced organismal lifespan (Bize *et al.*, 2009; Salomons *et al.*, 2009b). While the link between telomere dynamics and life expectancy is widely recognized, it remains unclear whether telomere shortening is a correlative and/or causative factor of mortality, and hence, inferences are often simplistic and controversial. Nevertheless, several lines of research suggest a strong connection between telomere dynamics and the processes that determine lifespan. *In vitro* studies demonstrate a clear link between telomere dynamics and cellular lifespan in the absence of telomerase (Allsopp *et al.*, 1992), while *in vivo* studies in identical twins demonstrate correlations between telomere dynamics and organismal lifespan (Bakaysa *et al.*, 2007). Likewise, mutations that reduce or silence the effect of telomerase genes result in greater telomere attrition, premature aging, and truncated lifespan (Blasco, 2007), and artificially increasing telomere length in nematodes extends lifespan (Joeng *et al.*, 2004). So, while much remains to be resolved, telomere dynamics nevertheless provide a useful tool for measuring the pressures that influence mortality, and such links provide a promising mechanistic basis with which to study hypotheses of aging and lifespan across disciplines.

Outside of the human paradigm, does sex have a role in telomere and lifespan shortening?

Women outlive men in every age group in nearly every population (e.g., 75% of the over 100s are women, Blagosklonny, 2010). This sexual inequality in lifespan is mirrored in the telomeres. At birth, there are no sex differences in telomere length (Table 2a), but thereafter, males tend to have shorter telomeres than females, indicating that male telomeres shorten faster (Table 2b; see Dialog S1; Fig. S2; Table S2, for an appraisal of the literature including a qualitative meta-analysis). Short leukocyte telomere length is correlated with some degenerative diseases (Table 2e), which are often more prevalent in males (Table 2f). Moreover, short leukocyte telomeres or faster rates of shortening are often (Table 2g), although not always (Table 2h), correlated with decreased survival in humans. So, while the direction of causality is not yet clear, it is tempting to describe the difference in telomere length or attrition rate as an underlying cause for the gender gap in longevity in humans and to extrapolate this reasoning to all vertebrates (e.g., Stindl, 2004). However, the link is nearly entirely based on human leukocyte research, and this has had major implications as to how sex-dependent telomere dynamics have been interpreted.

The relationship between sex, telomere shortening, and lifespan found in humans is not ubiquitous throughout taxa (Tables 1 and S1). Although non-human research is based largely on individual studies, often with a small sample size (See Table 1, Fig. 1), trends are nevertheless apparent. While male-biased mortality is often, although not always, accompanied

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Table 1 Summary of the relationship between sex and telomere dynamics in all studied taxa where sex was included in the analyses

Phylogenetic information			Telomere biology						Information pertaining to different hypotheses					Reference information	
Species/Population		Class	Sampled tissue[a]	Telomerase as adults[b]	Age-related telomere attrition	Smaller telomeres at start of life	Smaller telomeres as adults	Greater telomere attrition	Greater mortality	Heterogametic sex	Larger sex	Greater oxidative stress	Mating system[c]	Sample size	Telomere reference[f]
Human	*Homo sapiens*	Mammalia	WBC	×	✓	♂=♀	♂	♂	♂	♂	♂	♂	M	Varied	See Tables 1 and S1
House mouse	*Mus musculus*	Mammalia	WBC	✓	×	-	♂=♀	♂[1]	♂[2]	♂	♂	♂	P	48 45	Ilmonen et al. (2008) Kotrschal et al. (2007)
Brown rat	*Rattus norvegicus*	Mammalia	K, Li, L, P K	✓	✓	♂=♀	♂	♂	♂	♂	♂	♂	P	16 6	Cherif et al. (2003) Tarry-Adkins et al. (2006)
Algerian mouse	*Mus spretus*	Mammalia	K, Li, S,B	✓	✓	-	♂	♂	♂[3]	♂	♂	♂=♀	M	182	Coviello-McLaughlin & Prowse (1997)
Sand lizard	*Lacerta agilis*	Reptilia	WB	✓[4]	✓[5]	-	♂	♂[5] ♂[6]	♂	♂	♀	-	P	134 (9) 127	Olsson et al (2010) Olsson et al (2011)
Black garden ant	*Lasius niger*	Insecta	M, G, FE	✓[4]	✓[8]	♂=♀[7]	♂<♀w = ♀q[9]	♂[7]<♀w = ♀q[8,9]	♂<♀w < ♀q[9]	♂[10]	♀q > ♀w > ♂[9]	♂	C	<10	Jemielity et al. (2007)
Jackdaw	*Corvus monedula*	Aves	WB	-	✓	♂=♀	♂=♀	♂[12] ♂=♀	♂[13]	♀	♂	♂	M	385 & 136 166 & 44	Salomons et al. (2009a) Salomons et al. (2009b)
Cynomolgus macaque	*Macaca fascicularis*	Mammalia	M, K , Sk, S, Th, P, L, A	×[4]	✓[17]	-	-	♂=♀[18]	♂	♂	♂	♂	P	23	Gardner et al. (2007)
Northern giant petrel	*Macronectes halli*	Aves	WB	-	✓[11]	-	♂	♂=♀	♂	♀	♂	-	M	47	Foote (2008)
Southern giant petrel	*Macronectes giganteus*	Aves	WB	-	✓[11]	♂	♂	♂=♀	♂	♀	♂	-	M	63	Foote et al. (2011a)
Common tern	*Sterna hirundo*	Aves	WB	✓	✓	-	♂=♀	♂=♀	♂[13]	♀	♂=♀	-	M	44	Haussmann et al. (2003)[19]
Water python	*Liasis fuscus*	Reptilia	WB	✓[4]	×	♂=♀	♂	♂[14]	♀	♀	♀	-	P	70 & 13	Ujvari & Madsen (2009)
Blue footed booby	*Sula nebouxii*	Aves	WB	-	×[20]	-	♂=♀[20]	-	♀	♀	♀	-	M	69	Foote (2008)
Great frigate bird	*Fregata minor*	Aves	WB	-	✓	-	♂=♀	♂=♀	♀	♀	♀	-	M	33	Juola et al. (2006)
Zebra finch	*Taeniopygia guttata*	Aves	WB	✓	✓	-	♂=♀	♂=♀	♀	♀	♂=♀	♂=♀	M	30	Haussmann & Vleck (2002)
Sand martin	*Riparia riparia*	Aves	WB	-	✓[11]	-	♂=♀	♂=♀	♀	♀	♂=♀	-	M	20	Pauliny et al 2006
Leach's storm petrel	*Oceanodroma leucorhoa*	Aves	WB	✓	×	♂=♀	♂=♀	-	♀	♀	♂=♀	-	M	214 & 33	Haussmann et al. (2003)[19] Haussmann & Mauck (2008)[19]
Wandering albatross	*Diomedea exulans*	Aves	RBC	-	×	♂=♀	♂=♀	-	♀	♀	♂	♂=♀	M	92	Hall et al. (2004)
Adélie penguin	*Pygoscelis adeliae*	Aves	WB	-	✓ -	-	♂=♀	♂=♀	♀	♀	♂	♂=♀	M	22 72	Haussmann et al. (2003)[19] Beaulieu et al. (2010)
Kakapo	*Strigops habroptilus*	Aves	WB	-	✓[11]	-	♀	♂=♀	♀	♀	♂	-	P	69	Horn et al. (2011)
American alligator	*Alligator mississippiensis*	Reptilia	WB	✓[4]	✓	-	♂=♀	♂=♀	♀	ESD[21]	♂	-	P	26	Scott et al. (2006)
Chinese alligator	*Alligator sinensis*	Reptilia	WB	✓[4]	✓	-	♂=♀	♂=♀	♀[3]	ESD[22]	♂[3]	-	P[3]	30	Xu et al. (2009)
Humans (Amish)	*Homo sapiens*	Mammalia	WBC	×[4]	✓	-	♂=♀	♂=♀	♂=♀	♂	♂	-	M	907	Njajou et al. (2007)
European shag	*Phalacrocorax aristotelis*	Aves	WB	-	✓[11]	♀[15]	♂=♀	♂[16]	♂=♀	♀	♂	-	M	46 144 & 16	Foote (2008) Hall et al. (2004)
Dunlin	*Calidris alpina*	Aves	WB	-	✓	-	♂=♀ (but ♂[22])	♂=♀	♂=♀	♀	♀	-	M	7 & 28	Pauliny et al.(2006)
Black-tailed gull	*Larus crassirostris*	Aves	WB	-	✓	♂=♀	♂=♀	♂=♀	♂=♀	♀	♂	-	M	130	Mizutani et al.(2009)
Lesser black-backed gull	*Larus fuscus*	Aves	WB	-	✓[13]	♀	-	♂=♀	♂=♀ (or ♂[13])	♀	♂	-	M	23	Foote et al. (2011b)
Red bishop	*Euplectes orix*	Aves	WB	-	-	♂=♀	-	-	♂=♀	♀	♂	-	P	179	T. Friedl[23]
Alpine swift	*Apus melba*	Aves	WB	-	✓	-	♂=♀	♂=♀	♂=♀	♀	♂=♀	♀	M	96 & 22	Bize et al 2009
Seychelles warbler	*Acrocephalus sechellensis*	Aves	WB	-	✓	-	♂=♀	♂=♀	♂=♀[y]	♀	♂=♀	♂	M	12 & 400	Barrett et al. (In Review)
Tree swallow	*Tachycineta bicolour*	Aves	WB	✓	✓	-	♂=♀	♂=♀	♂=♀	♀	♂=♀	-	M	48	Haussmann et al. (2003)[19]

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Table 2 Summary of the relationships between telomere dynamics and survival in men and women. Please see Supporting Information for a qualitative meta-analysis

Phenomenon	Phenomenon description	References
a	No sex difference in telomere length at start of life	*In utero* (Youngren *et al.*, 1998); at birth (Okuda *et al.*, 2002)
b	Men's telomere shorten faster than women's	Reviewed by Stindl (2004) and Aviv *et al.* (2005), not just in old age (e.g. Mayer *et al.*, 2006)
c	Population differences. No sex difference in telomere attrition or lifespan in Amish.	Njajou *et al.*(2007)
d	Aging associated with telomere attrition	Reviewed in Sahin & DePinho (2010)
e	Disease states related to short telomeres and increased attrition rates Cardiovascular disease Dementia Diabetes Liver cirrhosis Ulcerative colitis Renal failure	Reviewed in Blasco (2005)
f	Men have greater susceptibility to degenerative disease	Relationship between cardiovascular disease and sex, reviewed by Aviv (2002)
g	Short telomeres or greater telomere attrition associated with decreased survival	e.g. Bakaysa *et al.* (2007), Njajou *et al.* (2007), Kimura *et al.* (2008), Atzmon *et al.* (2010)
h	Shorter telomeres not found to be related to survival	e.g. Martin-Ruiz *et al.* (2005), Harris *et al.* (2006)

by shorter male telomeres or a faster attrition rate, no studies have reported greater telomere shortening in females, including situations where females have greater mortality. Moreover, in species where females have shorter lifespans, males can still have greater telomere shortening (Table 1). So then what is the cause for the sexual dimorphism in telomere dynamics and lifespan? Below, we review the role of sex in telomere dynamics across species to investigate whether this can provide a new perspective on sex-dependent longevity phenomena. We consider four areas where significant information has been obtained on the role of sex in telomere biology: (i) genetic heterogamety, (ii) body size, (iii) oxidative stress, and (iv) telomere maintenance.

The heterogametic disadvantage

In birds and mammals, adult mortality tends to be higher in the heterogametic sex (Liker & Szekely, 2005), i.e., the male in mammals (XY) and the female in birds (ZW). In mammals and XY reptiles, the sex difference in lifespan is generally accompanied by differences in telomere attrition (Table 1). However, sex differences in telomere dynamics are rare in birds, ZW reptiles, and reptiles with environmental sex determination, and despite female-biased mortality being the norm in these taxa, greater telomere attrition in females has yet to be described (Table 1).

Theory suggests that the heterogametic sex suffers greater mortality because of the unguarded expression of any deleterious recessive alleles residing on their sex chromosomes (Liker & Szekely, 2005). The competi-

Please see Table S1 (Supporting information) for details on experimental detail and reference for mortality and telomerase data. Blue = male bias; pink = male bias; yellow = no sex difference; grey = no information; brown = mammal; orange = bird; green = reptile; purple = insect.
A, adipose; B, brain; FE, fertilized eggs; G, gonads; K, kidney; L, lung; Li, liver; M, muscle; P, pancreas; S, spleen; Sk, skin; Th, thyroid; WB, whole blood; WBC, white blood cells; RBC, red blood cells; N, down-regulated (low levels or nil); Y, active; M, monogamous (includes socially monogamous); P, polygamous; C, colonial; L, longtiudinal analysis (> 1 sample per individual – within individual changes); ×, cross-section analysis (one sample per individual – between individual trends).
Superscript numbers refer to: [1]Under stress (parasite or social). No age related telomere decline in control conditions; [2]While several studies show shorter ♂ lifespan (e.g. refs), sex difference may vary (reviewed in Austad 2011). Lifespan data is not from the same study as telomere data; [3]Data is not conclusive; [4]Not based on this species, but on the general finding that telomerase activity is found in this class (Gomes *et al.* 2010); [5]Large ♂ size and tail regeneration associated with small telomere length in ♂ but not ♀; [6]Sex dependent, although data (which suggests ♀ shortening) does not seem consitent with interpretation (which suggests ♀ main telomere length through life better than ♂ do); [7]N=1; [8]Based on comparisons of fertilized eggs and somatic adult tissue rather than in adult life; [9]☿w = worker; ☿q = queen; [10]♂ haploid, ♀ diploid; [11]Adult telomere lengths shorter than chicks, there is no age-related decline in telomere length observed in adults; [12]♂raised in enlarged broods have greater telomere shortening; there is no effect of brood manipulation on ♀; [13]Juveniles; [14]Hatching telomeres shorter than adults, data suggest greater ♀ lengthening opposed to greater ♂ attrition; [15]Based on work by Foote (2008); Hall *et al.* (2004) found no difference, but had a wider age range of chicks; [16]♂ have shorter telomeres than ♀ as young chicks, this difference is lost, suggesting greater ♂attrition in juvenile period; [17]In skin, thyroid and lung; [18]Only give statistic for muscle; [19]Pers. comm. with M. Haussmann based on data from referenced article; [20]Results based on flawed methodology; [21]Environmental sex determination; [22]Body size has negative correlation with telomere length in ♂, but not ♀; [23]Average telomere length in ♂ nestlings not significantly different from ♀ nestlings; t-test, p = 0.637 (Thomas Friedl pers. comm.).
*Telomerase activity can vary between tissues and with age. Telomerase expression is given for the sampled tissue, unless otherwise stated.
†Reference for sex-related telomere dynamics only.

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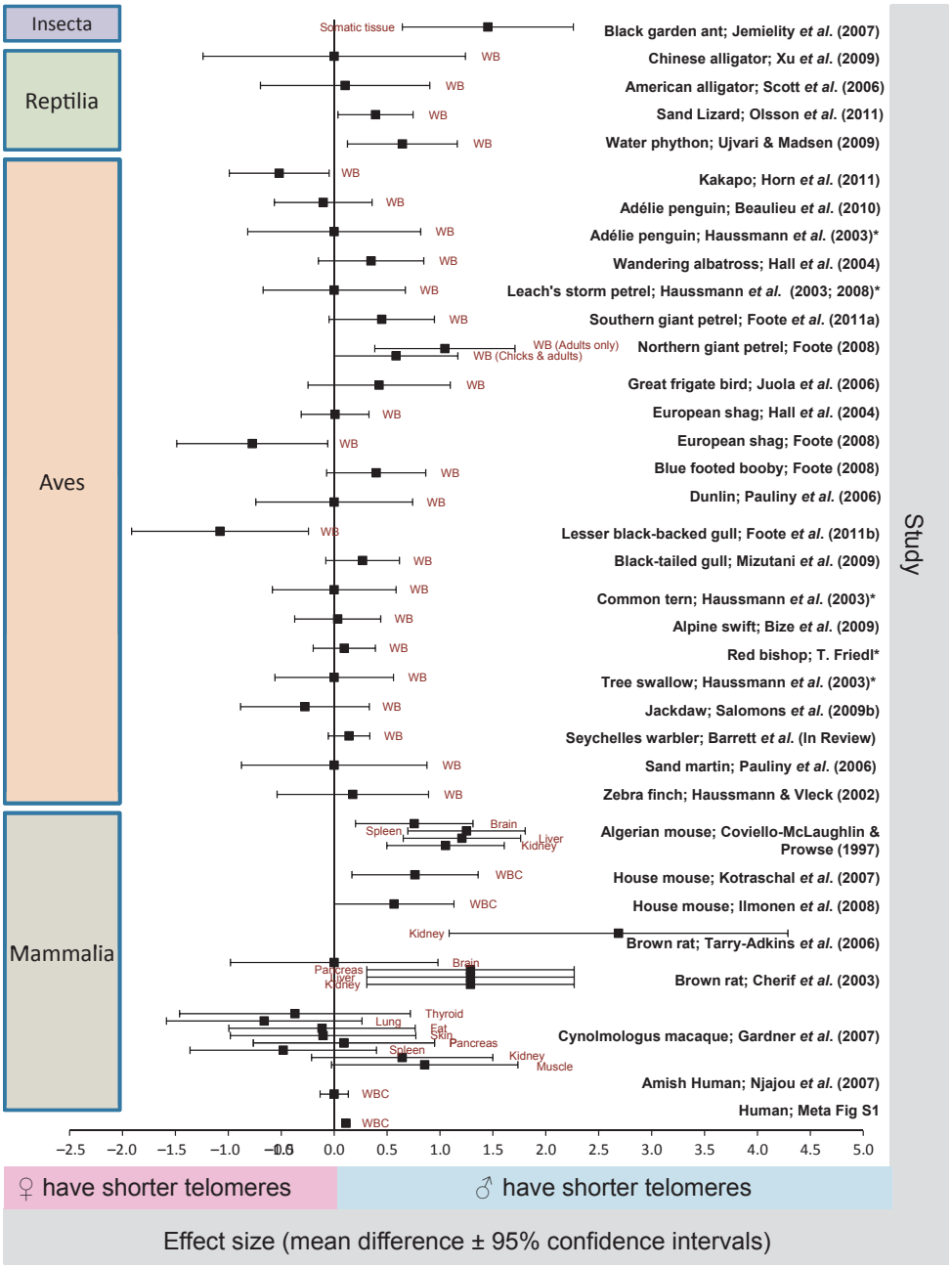


Fig. 1 Forest plot diagram of studied taxa from Table 1. Squares connote effect sizes for sex difference in telomere length only (i.e., not telomere shortening) ± 95% confidence intervals. The central vertical line represents a mean difference of zero (the null hypothesis of no sex difference in telomere length). When the error bars to not include this line the effect is significant. Note that while there is an apparent general bias for females to have longer telomeres, this is rarely significant in birds compared to mammals, although larger sample sizes are required to reduce error.

tive advantage to the homogametic sex can be plainly seen at a cellular level in eutherian mammals because of stochastic X-inactivation. To prevent XX females from having a double dose of X gene product compared to XY males, one X chromosome is stochastically inactivated in each cell in the female embryo (Lyon, 1961). Subsequently, female neonates are a 1:1 mosaic of two active X-cell populations. Over time, the ratio becomes skewed to favor one population with the active X that, presumably, contains alleles that confer a cellular survival advantage (Hemizygous selection; Abkowitz *et al.*, 1998). Women over 60 often have leukocyte ratio skews of 3:1 (Busque *et al.*, 1996). Intriguingly, the cell lineages that have greater survival also have an active X chromosome with longer telomeres than the inactive X (Surrallés *et al.*, 1999), suggesting the survival advantage could be the result of variation in telomere maintenance alleles on the X chromosome. Males on the other hand acquire one X in addition to an unguarded Y. If the unguarded sex chromosomes contain defunct or inferior telomere maintenance alleles, they may contribute to the

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difference in telomere shortening between sexes. Haplodiploid species (where males develop from unfertilized haploid eggs) may prove an extreme example of this, as males only have one copy of all potential telomere maintenance genes, whereas females have two. Interestingly, in black garden ants (*Lasius niger*), males have shorter somatic telomeres than females and shorter lifespans (Jemielity *et al.*, 2007). However, what makes the social hymenoptera a particularly fascinating model of longevity is the segregation of females into queens and workers. Queen black garden ants can live up to 28 years, while female workers live only a few days; yet, there are no differences in telomere dynamics between the classes of females (Jemielity *et al.*, 2007).

For sex differences in telomere maintenance to arise from sex chromosome haploidy, genes involved in telomere maintenance would have to sit on the sex chromosomes. In humans, genes influencing telomere length have been mapped to autosomal (Andrew *et al.*, 2006) and sex chromosomes (Connor *et al.*, 1986). A mutation in the *DKC1* gene on the X chromosome involved in telomerase function causes X-linked dyskeratosis congenital, a disease characterized by rapid telomere attrition, accelerated aging, and short lifespan (Bessler *et al.*, 2004). Sex differences in telomere attrition are rare in birds, and there is no homology between the mammal XY and bird/reptile ZW chromosomes. If telomere maintenance genes are autosomal, rather than sex-linked in birds, we may not see avian sex differences in telomere attrition. Yet, there still may be sex differences in lifespan because of other deleterious recessive alleles on sex chromosomes. This could explain female-biased mortality without female-biased telomere attrition in birds and ZW reptiles (Table 1). Although the *DKC1* gene is located on chromosome 4 in birds, there is still scope for sex differences in telomeres because of heterogamety in ZW taxa as other telomere maintenance genes are located on the sex chromosomes (e.g., cPOT1 on the Z of the chicken; Wei & Price, 2004).

A related theory is that sex-specific parental imprinting of telomere maintenance genes could drive sex-linked telomere maintenance. How a gene behaves can depend on the sex of the donor parent, with different consequences for offspring of each sex (Wijchers & Festenstein, 2011). Heterogamety could lead to a mismatch if paternal imprinting controls expression of telomere maintenance genes in one direction and maternal imprinting works in the reverse (Njajou *et al.*, 2007). Conceivably, this may lead to the down-regulation of telomere maintenance genes in the heterogametic sex, with subsequent telomere shortening. In humans, where men are the heterogametic sex, it appears that there is a stronger father–child relationship in leukocyte telomere length, compared to a weak or insignificant mother–child relationship (Nordfjäll *et al.*, 2005, 2010; But see, Akkad *et al.*, 2006). Conversely, in birds, where the females are the heterogametic sex, the inheritance trend is reversed and inheritance is characterized by a stronger mother–child relationship in erythrocyte telomere length, but no relationship between father and offspring (Jackdaws (*Corvus monedula*), Salomons *et al.*, 2009a; Kakapo (*Strigops habroptilus*), Horn *et al.*, 2011). Horn *et al.* (2011) propose that the sex chromosomes regulate imprinting of autosomal genes, either directly through genes located on the Y chromosome in humans and the W chromosome in the kakapo or indirectly through expression levels on the X and Z, respectively. However, while male kakapo erythrocyte telomere lengths are longer than those of females (Fig. 1), there is no apparent telomere shortening with age in either sex (from sub-sample of lifespan, Horn *et al.*, 2011). Longer male erythrocyte telomeres have also been found in lesser black-backed gull (*Larus fuscus*) chicks, but similarly, telomere attrition rates do not differ between the sexes post-hatching (Fig. 1; Foote *et al.*, 2011b). Therefore, while the combination of imprinting and heterogamety may be important in the inheritance of telomere length in birds, its relevance to telomere maintenance is questionable.

Bigger not always better?

Sex-biased mortality is more pronounced in species that display sexual size dimorphism (Clutton-Brock *et al.*, 1985). To attain and maintain a larger body size, it is argued that the cells of the larger sex undergo more cellular replication and will lose more telomere to inefficient DNA replication; hence, sexual size dimorphism could explain dimorphism in telomere attrition and potentially lifespan (Stindl, 2004). Among mammalian species, large body size is also associated with reduced telomerase expression – a predicted response to the greater risk of cell immortalization accompanying greater cell number (Seluanov *et al.*, 2008; Gomes *et al.*, 2011). Could the influence of sexual size dimorphism have resulted in differential telomerase expression between the sexes? At a cellular and organismal level, it appears that size could be a problem, if only for males. When males are larger than females, they frequently have shorter telomeres, whereas monomorphic species rarely have any sex differences in telomere loss (Table 1). Moreover, one cell culture study showed that donors who were tall men had lower cell replicative capacity than shorter men (Maier *et al.*, 2008). But large male size is not consistently accompanied by greater telomere loss (Table 1) and in some species females are the larger sex, and yet, greater female telomere shortening has yet to be documented. Indeed, in some cases, males still have greater telomere attrition (Table 1).

Telomerase activation is tightly linked to growth (Greider, 1998), and in the few cases where it has been studied, females have greater telomerase activity (Leri *et al.*, 2000). If growth-dependent telomerase activation differs between the sexes, we might expect to find size-dependent telomere attrition in one sex, but not the other. The Maier *et al.* (2008) study, discussed earlier, showed that human cell replicative capacity is dependent on height when the donor is male, but not when female. Moreover, large male dunlins (*Calidris alpine)* and sand lizards (*Lacerta agilis*) have shorter telomeres than small males, but size does not affect female telomere length, despite them being the larger sex (Pauliny *et al.*, 2006; Olsson *et al.*, 2010; respectively). Nevertheless, most species with no sex differences in telomere length or shortening do still show an age-related decline in telomere length (Table 1), which we might not expect if telomeres were fully maintained by telomerase.

The oxidative stress of being big, hormonal, and riddled with disease

The cumulative damage to biomolecules caused by oxidative stress is a popular candidate for explaining aging and lifespan (Møller *et al.*, 2010). The nucleobase guanine, prevalent in the telomeres, is particularly sensitive to oxidation by ROS (Oikawa & Kawanishi, 1999). Furthermore, while breaks in the internal genome DNA are repaired quickly, repairs to equivalent breaks in telomeres are slow and incomplete (Petersen *et al.*, 1998). Consequently, oxidation by ROS could induce considerable telomere attrition and cellular senescence (Liu *et al.*, 2002). Sex hormones can interact with ROS production and management. Estrogen reduces the production of ROS while also being a potent antioxidant and regulator of antioxidant genes (Viña *et al.*, 2005). Conversely, testosterone has no antioxidant properties and is linked to increased susceptibility to oxidative stress (Alonso-Alvarez *et al.*, 2007) and reduced immunocompetence (Muehlenbein & Bribiescas, 2005). Immunocompetence is further linked to telomere attrition as the innate immune system generates ROS to fight parasites (Bogdan *et al.*, 2000), which may cause oxidative stress and telomere damage. For example, male house mice (*Mus musculus*) are more susceptible to *Salmonella* infection than females, and despite adult somatic telomerase activity in both sexes (Prowse & Greider, 1995),

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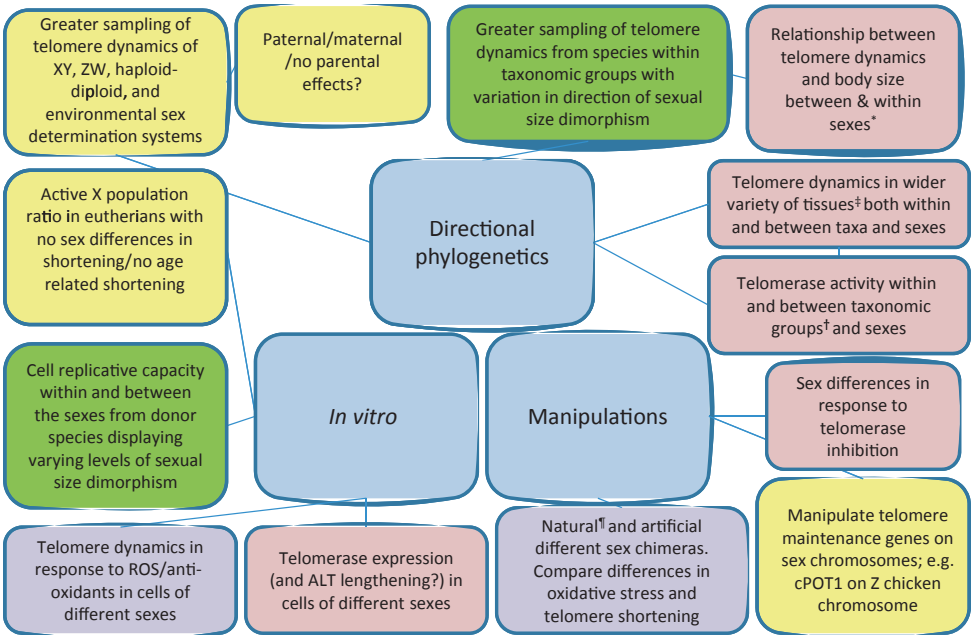


Fig. 2 Future directions for sex-specific telomere biology. Colours represent research aims in the four main areas (i) genetic heterogamety (yellow), (ii) body size (green), (iii) oxidative stress (purple), and (iv) telomere maintenance (pink). *To determine whether being larger sex is only a problem if it is the male. †As in Seluanov *et al.* (2008) in rodents and Gomes *et al.* (2011) in mammals, but applied to the sexes. ‡Telomere dynamics are not consistent between tissues of an individual. ¶More common in non-human taxa.

leukocyte telomere attrition is affected by infection in males, but not in females (Ilmonen *et al.*, 2008).

Despite the links between sex hormones and parameters of oxidative stress in mammals, equivalent sex differences metabolic rate, ROS production, and antioxidant capacity are not common in birds (Cohen *et al.*, 2008). It may, therefore, be tempting to link a relative lack of sex differences in oxidative stress to a lack of dimorphism in telomere attrition in birds. Yet, across all taxa, there are many specific examples that illustrate that the relationship between sex, oxidative stress, and telomeres is not that clear. For example, male and female Algerian mice (*Mus spretus*) have equal levels of oxidative stress (Bonilla-Valverde *et al.*, 2004), but males have greater telomere shortening in many tissues (Coviello-McLaughlin & Prowse, 1997). Whereas female alpine swifts (*Apus melba*) and Seychelles warblers (*Acrocephalus sechellensis*) have more and less oxidative stress than males, respectively (Bize *et al.*, 2008; van de Crommenacker *et al.*, 2011), yet there are no sex differences in erythrocyte telomere shortening (Bize *et al.*, 2009; Barrett *et al.*, unpublished data). Clearly, the link between oxidative stress and telomere shortening *in vivo* remains equivocal and requires further study.

Sex differences in telomere addition

Telomere dynamics are influenced by the loss and gain of telomeric base pairs. While there are many reasons why males may lose more telomere than females, there are also reasons why females may gain more than males. Sex differences in lifespan are often explained as a consequence of the differential costs of reproduction (Vinogradov, 1998). Sperm is cheap, but competition for the female's expensive reproductive investment is fierce and can encourage risky male behavior. Under the disposable soma theory, this can lead to sexual selection on patterns of senescence (Bonduriansky *et al.*, 2008), whereby higher mortality in one sex reduces investment in somatic maintenance (e.g., telomeres; Jemielity *et al.*, 2007) and shortens intrinsic lifespan (Kirkwood, 1990). Sex-specific risks are less clear in birds, where females are frequently exposed to greater extrinsic risks than males, often resulting in female-biased mortality (Liker & Szekely, 2005), but we still do not observe greater telomere shortening in females.

The influence of sex on telomere maintenance is only just beginning to be explored, but findings suggest that a sex bias in telomere maintenance does exist. Estrogen directly activates a promoter of telomerase (Kyo *et al.*, 1999) and indirectly also affects DNA repair through the p53 pathway (Sengupta & Wasylyk, 2004) and telomerase activation through the phosphoinositol-3-kinase/Akt (Simoncini *et al.*, 2000) and nitric oxide pathways (Grasselli *et al.*, 2008). These mechanisms result in females having greater telomerase activity in some species (e.g., rats; Leri *et al.*, 2000). Consequently, greater female longevity may be due to greater telomere addition, secondary to estrogen exposure. Indeed, post-menopausal women on hormone replacement treatment have longer leukocyte telomeres than untreated women (Lin *et al.*, 2010). Androgens can also have similar effects on telomerase in the presence of aromatase (which converts androgens to estrogens), and these can be used to treat telomerase failure syndromes (Calado *et al.*, 2009). However, even untreated post-menopausal women have slower leukocyte telomere attrition than age-matched men (Mayer *et al.*, 2006), despite men having more estrogen at this stage through testosterone conversion. This may be because the estrogenic control of telomerase activation is not the sole mechanism females have for lengthening telomeres. Möller *et al.* (2009) uncovered the evidence for female utilization of leukocyte telomere lengthening by recombinant sister chromatid exchange. We now need to investigate whether this mechanism is under the control of sex hormones and whether being the 'risk adverse' sex could have lead to the evolution of sex-specific alterative pathways of telomere maintenance in mammals.

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Conclusions and future directions

Despite the various candidate theories for sex differences in telomere maintenance and lifespan, none can independently explain the diverse relationships between sex, telomeres, and lifespan observed across taxa. This finding reconfirms the dangers of making generalizations based on correlations observed in single taxon (i.e., in humans). Nevertheless, human X-linked diseases show that genetic heterogamety can play a role in telomere maintenance, while the internal sex environment also appears to be important. For example, Brüderlein *et al.* (2008) studied a pair of dizygotic twins chimerical for haematopoietic stem cells (cells from one twin in the bone marrow of the twin of opposite sex and *vice versa*). They found that male-derived leukocyte telomeres were 33% longer in the female than in the male, while female-derived leukocyte telomeres were 87% shorter in the male than in the female. Although only a single case study, this work suggests the sex-specific biological environment has considerable influence over telomere dynamics. The numerous studies on human leukocytes have provided the groundwork to enable us to ask informed questions, but we reiterate to medical researchers the importance of not making general conclusions based on one taxonomic group. We also emphasize to those researchers working with other organisms the importance of appropriate sample size, as valid conclusions can be limited by small sample sizes. In Fig. 2, we suggest areas for future research to help illuminate the role of sex in telomere biology. Such studies are essential to understand whether shorter telomeres in a variety of tissues are simply symptomatic of reduced survival or play a causal role: not just between the sexes, but within life as a whole.

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Supporting Information

Additional supporting information may be found in the online version of this article:

Fig. S1 Forest plot of literature on sex and telomere length in humans.

Fig. S2 Funnel plot of spatial heterogeneity of reporting of sex and telomere length in humans.

Table S1 Extended version of Table 1 of the main document including experimental information and further references.

Table S2 Output of quantitative meta-analysis models for the relationship between telomere length and sex.

Dialog S1 Investigating the assumption; do women have longer telomeres than men?

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